

Installation and Operating Recommendations

Balanced Weave Belts

The Manufacturer guarantees quality of workmanship and the metal used in the fabrication of the belt to be as specified. Belts are tested for true travel and rigidly inspected before shipment.

The care used in the installation of the new belt and the observances of good operating practices are insurance of long operating-life and low-maintenance cost. The following suggestions are offered for your guidance.

GENERAL OVERVIEW

This system consists of three (3) Balance Weave Bake Belt Bands. Air Turn is designed to take the product from the Top to Middle Baking Levels; Stationary Slide from Middle to Bottom Baking Levels; Belt Guide Rollers and Counter-weight Belt Tensioning and Tracking System is designed for each belt. The oven Infeed Conveyor is a full-width Balance Weave wire mesh belt with a magnetic roller drive system

Pre-Operations Check

Belt tension beyond that which is required to move the loaded belt should be avoided. Where conditions permit, the drive pulley should be lagged. Snub rolls, pressure or pinch rolls effectively reduce belt tension and are in general use on Wire Belt installations.

Terminal pulleys should be adjusted to be square to the conveyor centerline. To correct belt side travel re-check terminal alignment and make corrections to the belt supports as necessary. Permanent damage to the belt can occur if terminals are “cocked”. See Bulletin IC75 for additional information on belt tracking.

Supporting rolls must be level and at right angles to the conveyor center line. Belt supporting surface should be checked for freedom from obstructions such as warped hearth plates or protruding framework, which might cause belt impingement. Check the belt take-up mechanism to make certain it is functioning properly.

Before installing the new belt, make sure none of the spirals have been turned during shipment and that all spirals are flat and free to turn on their proper axis

Installation and Operating Recommendations

Balanced Weave Belts

TO MAKE BALANCED WEAVE BELTS ENDLESS (Refer to Sketch)

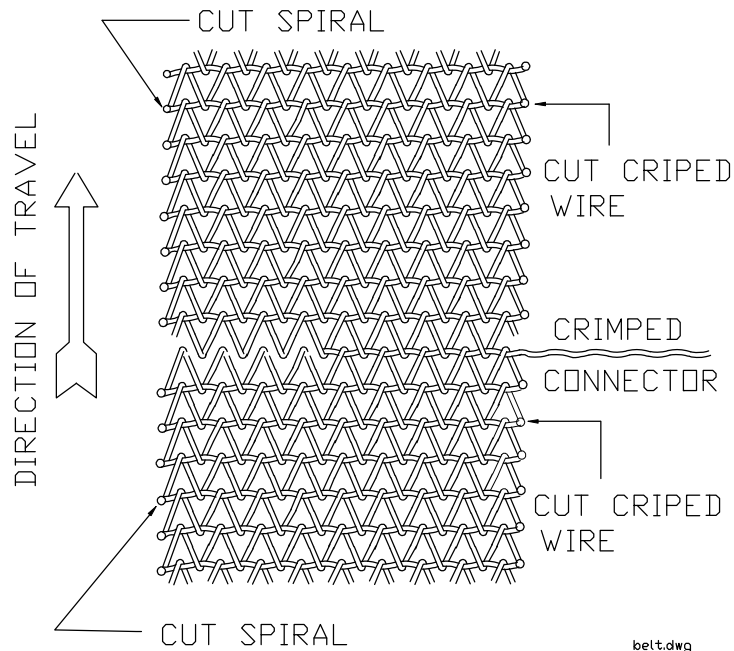
For ease of handling, most belts, because of their length and weight, are shipped in two or more sections. Normal care should be used in uncrating to prevent damage to the fabric.

The same method of joining the sections applies to making the bed endless. Our practice is standard whether the belt is shipped in one continuous length or is made up of two or more sections. All belt lengths terminate with a loose spiral. The loose spiral is left hand (counter clockwise windings). The fixed spiral that is welded to the crimp rod at the opposite end is right hand (clockwise winding).

Extra crimped connectors and spirals are included with every bed shipped. The spirals have a long end loop to facilitate fixation and the crimped connecting wires are longer than required for the belt width.

TO MAKE ENDLESS-bring the two sections together with the edges of the bed in line. Mesh the end spirals together to permit insertion of the crimped connector wire. Make certain the spirals so joined are of opposite weave-one right, the other left hand.

After inserting the connector wire, close down or knuckle the open ends of the spiral. Cut the crimped connector to the proper length and weld or turn in the protruding ends of the connector.



Installation and Operating Recommendations

DIRECTION OF BELT TRAVEL (Refer to Sketch)

When installing the belt, be sure that the spiral LEADS are crimped and attached at both ends of the rod.
Caution: Before tension is applied see that all spirals bottom or seat on the connector and lie flat.

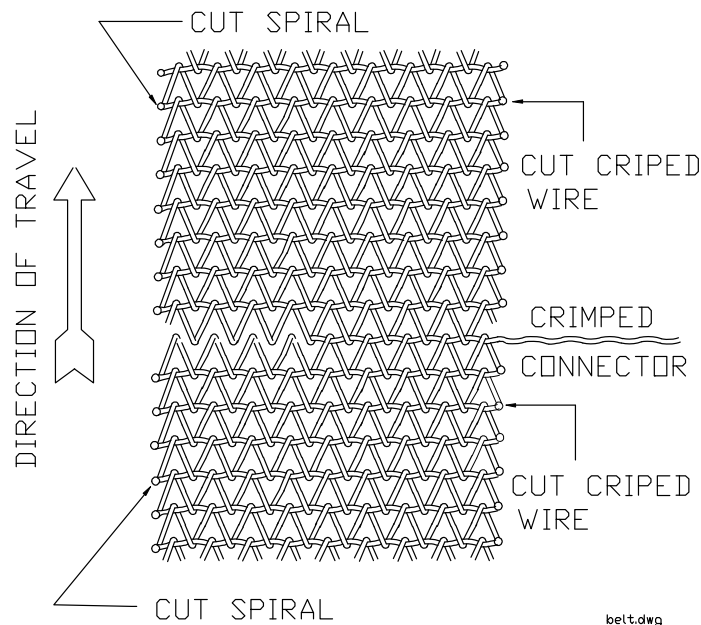
TO REMOVE STRETCH FROM BALANCED WEAVE BELTS

Normally, maximum stretch takes place during the initial period of operation. When the stretch exceeds that which can be taken up by the TAKE-UP device, a section of belt must be removed. Tag the section to be removed, take out as much as possible without making the re-assembly too difficult.

Follow procedure shown in the sketch. Inspect the spirals that were cut to remove the section are both of the same “turning”, (i.e., both right or both left hand spirals). On one side, cut the crimped wire, on the opposite side cut the spiral to which the crimped wire is fastened (welded). Then pull out the crimped connector wire and next turn out the shortened spiral.

The belt is now ready to be made endless again. Turn in a new spiral of the same wind as the one removed. Bring the two sections together with the edges of the belt in line. Mesh the end spirals together and insert the crimped connector wire. Make certain the spirals so joined are of opposite weave, one right hand and the other left hand.

Close down or knuckle the open ends of the spiral. Cut the crimped connector wire to the proper length and weld or turn in the protruding ends of the connector.



Installing And Adjusting New Belts

NOTE: Two-belt slide rail systems are used on each belt. One system supports the top or product carrying portion and the other supports the lower or return portion.

Both types of belt slides should be adjusted so that the belt drops 1/8" to 1/4" between where it leaves the drum and contacts the slide.

Mach IV ovens have a chevron rail system to guide the belt. The procedure for aligning the chevrons for the new belt is as follows:

1. Tie a nylon fishing line tightly around the front and rear drums as shown in figure 1. Make sure to do the adjusting procedure at A, B, C, D, E, and F as shown in figure 2.

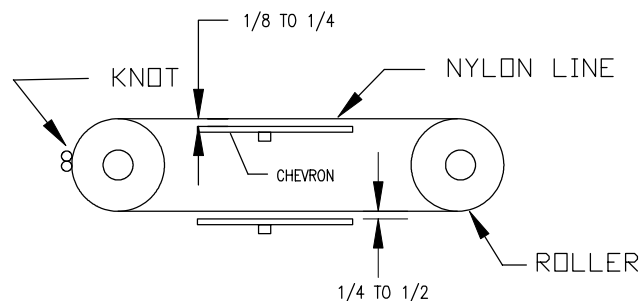


FIGURE 1

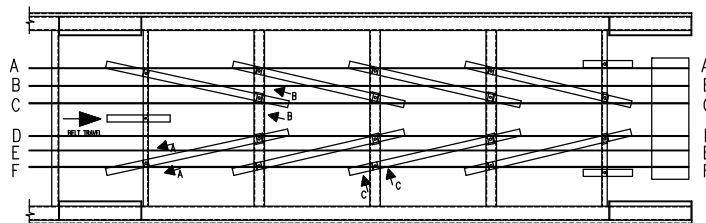


FIGURE 2

2. Adjust lower rails 1/8" to 1/4" below nylon lines. If lower than given figures add shims to the chevron bases as shown figure 3.

Installing And Adjusting New Belts

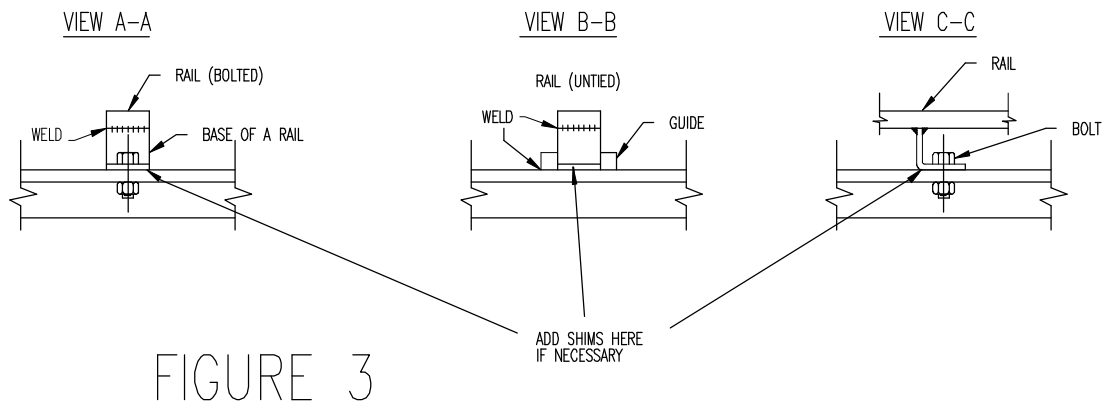


FIGURE 3

3. Insert “bubble” type level between rails and nylon lines and level rails across oven
4. Adjust upper rails 1/4” to 1/2” below nylon lines. If lower than given figures add shims to the chevron bases as shown in figure 3.
5. Insert “bubble” type level between rails and nylon lines and level rails across oven.

PARALLELING DRUMS OR ROLLERS

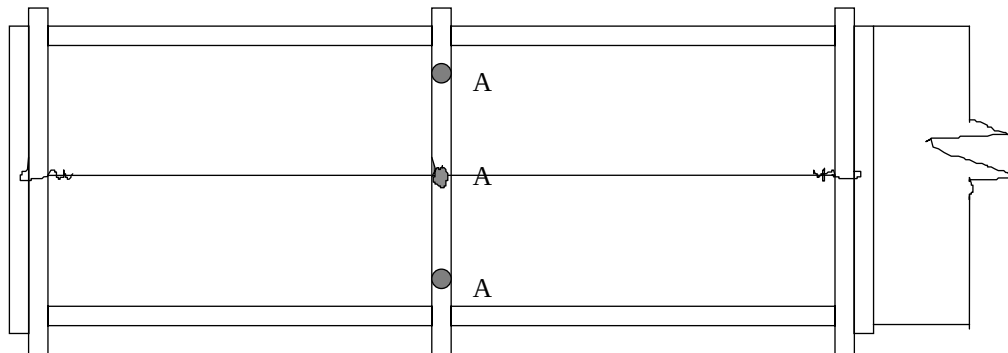
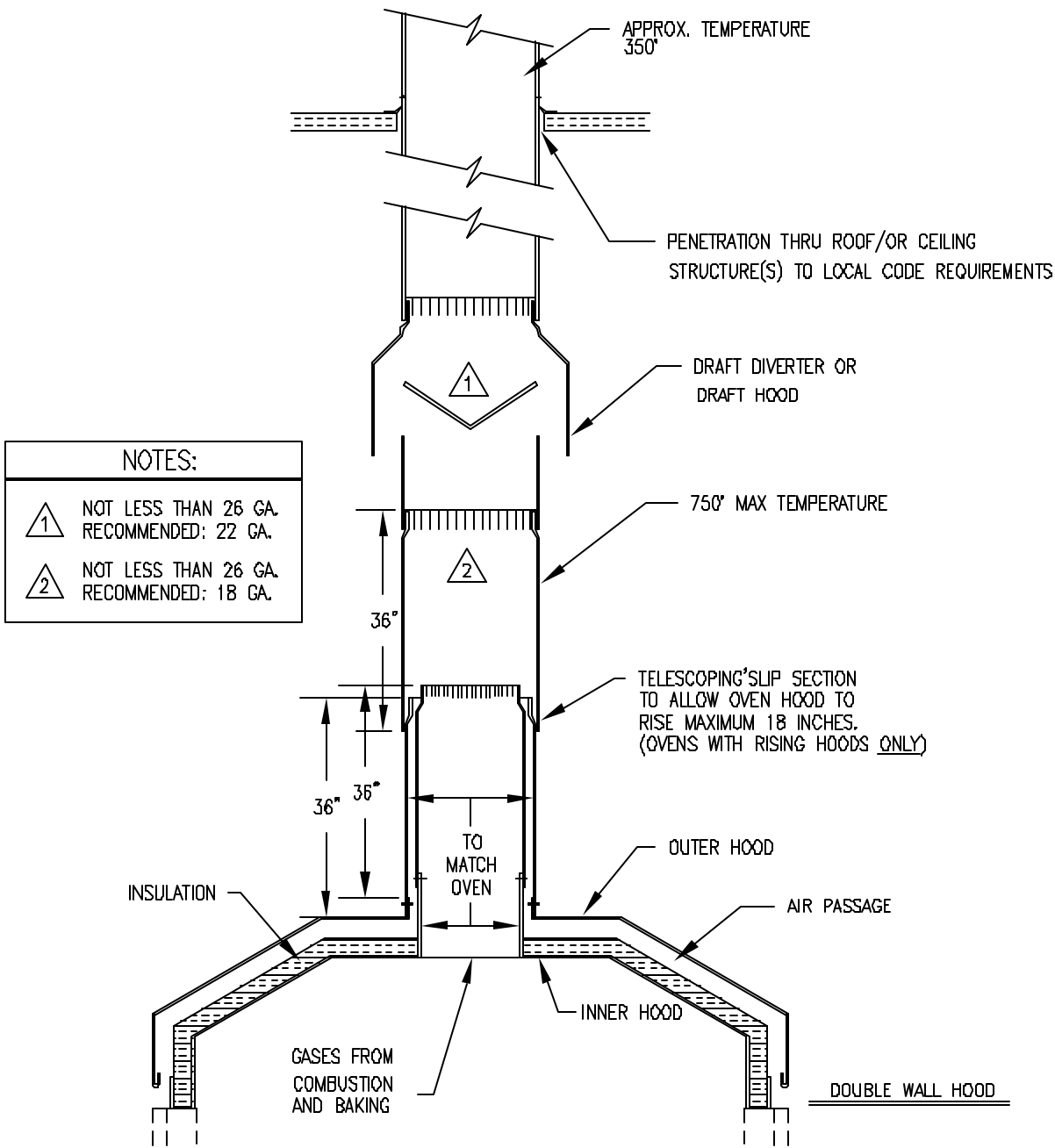


Figure 5

Installing And Adjusting New Belts

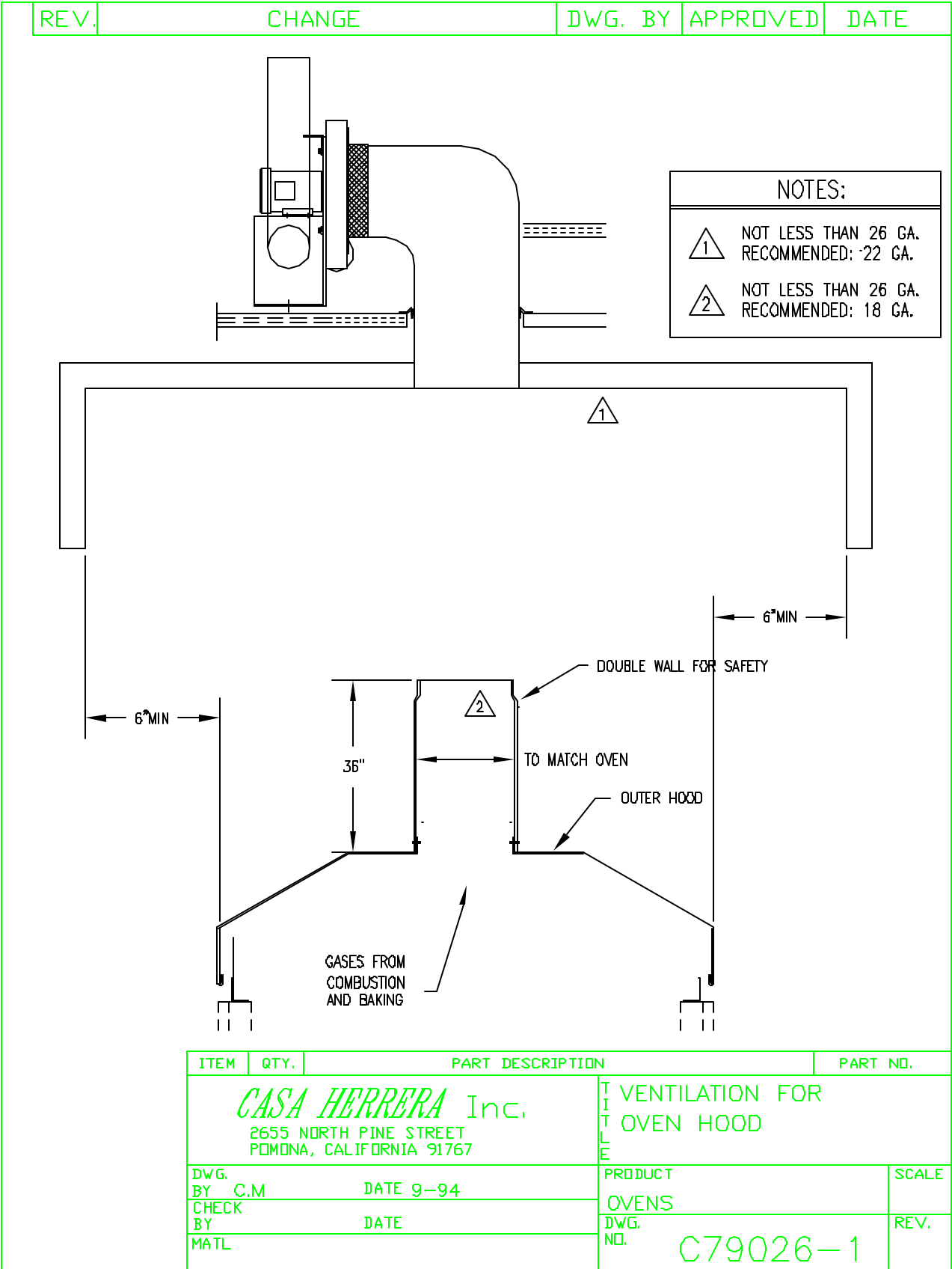
Paralleling Drums Or Rollers

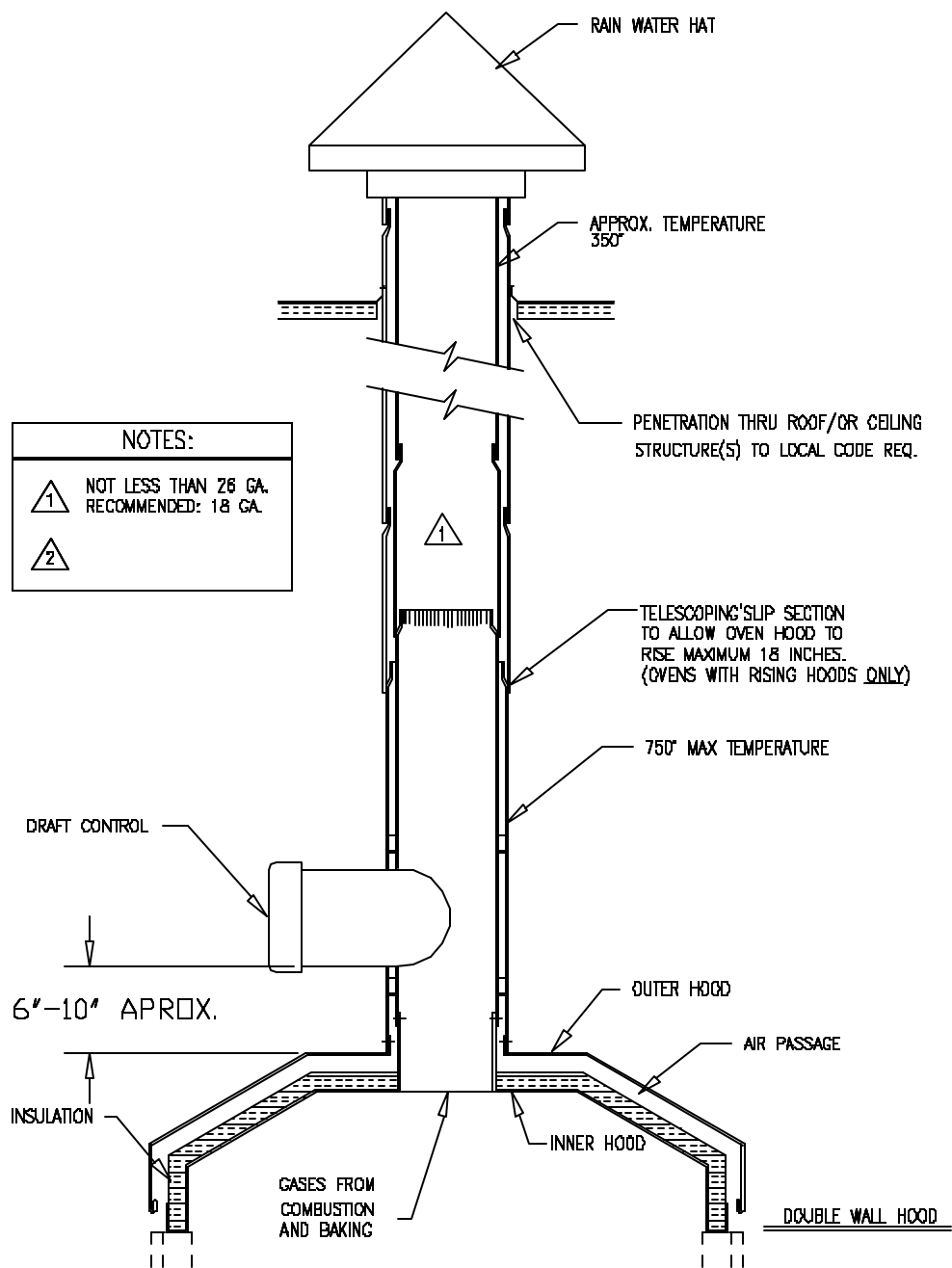
- ❑ Clamp a third metal straight edge or flat bar across the oven roughly midway between the two drums, and mark where the centerline of the oven intersects the approximate centerline of the straightedge or flat bar (see figure 5).
- ❑ Use a center finder (e.g. Racine “Mercury Balanced Miracle Point”), mark each end of each drum at “top dead center” and 1/2” in from each end. Check that distance from mark to centerline of oven is the same for each side of drum.



CHINS 0068

Herrera, Inc. 3860 S. METTLER ST. LOS ANGELES CA 90003			T- I T L E VENTING DETAILS FOR OVEN HOODS	
DWG. BY C. MEZA	DATE 9-84	PRODUCT	SCALE	
CHECK BY CDK	DATE 9-21-95	DWG NO. C79026	REV. B	
MATL				





CHNS0068

<i>CASA HERRERA</i> Inc. 5860 METTLER STREET LOS ANGELES, CALIFORNIA 90003		TITLE VENTING DETAIL FOR OVEN HOOD	
DWG. BY R.ROMERO	DATE 8/8/96	PRODUCT	SCALE
CHECK BY	DATE	DWG. NO. C79026-2	REV. C

Oven troubleshooting

The oven doesn't turn on.

1. Make sure that there is no one working in or around the oven. Make sure that no one has their hands or tools in the oven.
2. Must check that the oven panel main disconnect to make sure that it is on and that there is power to the panel (Figure 3.1)



Figure 3.1



Figure 3.2



Figure 3.3

3. Must be sure that all emergency stops are disengaged at the four corners of the oven (Figure 3.2).
4. Must check that the local disconnects for each motor are turned on (Figure 3.3).
5. You must check that there is gas pressure to the oven (Figure 3.4).



Figure 3.4

6. You must check that there is combustion air from the combustion blower to the oven (Figure 3.5).



Figure 3.5



Figure 3.6

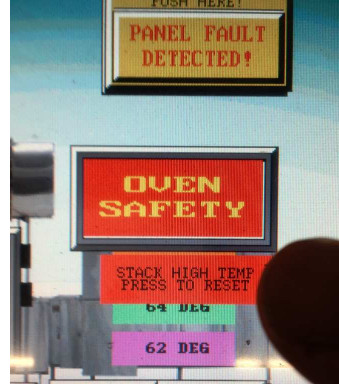


Figure 3.7

7. Also check that there is compressed air going to the oven if needed (Figure 3.6).

8. Reset the 'Stack Hi-limit alarm' by holding down the reset button on the touchscreen. Every time the power is turned off at the panel, the 'Stack Hi-limit' switch has to be reset (Figure 3.7).

9. Turn 'Oven Belts' on (Figure 3.8).

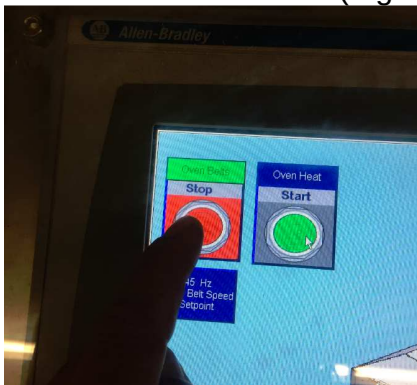


Figure 3.8

10. Turn 'Combustion Blower' on.

11. Turn the 'Heat' on and wait 30 – 90 seconds. The pilot gas solenoid valve will open to feed fuel to the pilot. The 110 volt AC transformer will convert electricity to 6,000 volts DC for the igniter. ***Warning: Do Not Touch the Igniter during this Process*** This spark will light the fuel in the pilot tree (Figure 3.9 & 3.10).



Figure 3.9



Figure 3.10

If there is no spark.

1. Check to see that the rubber insulated igniter cable is not damaged beginning at the transformer and ending at the igniter. If there is a crack or a broken insulation area in the igniter cable, then the electricity will not reach the igniter and will go to 'ground' before it reaches the fuel coming out of the pilot tree (Figure 3.11).



Figure 3.11

2. If the spark is ok, then check to see if you have power at the terminal strip for the pilot gas solenoid valve (Figure 3.12). If this valve is receiving power and opening, then check the air/gas mixture for the pilot flame and combustion.

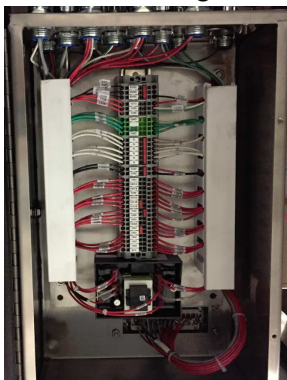


Figure 3.12



Figure 3.13

3. If you get a solid flame on your pilot tree, and the main burner does not light, then you need to check the 'flame strength signal' at the burner controller (Figure 3.13).

4. There is a U.V. scanner looking at the pilot tree flame (Figure 3.14). If the U.V. scanner is obstructed and cannot see the flame, then the main gas valve will not be allowed to open. Make sure that there is nothing obstructing the view to the pilot tree (Figure 3.15).



Figure 3.14

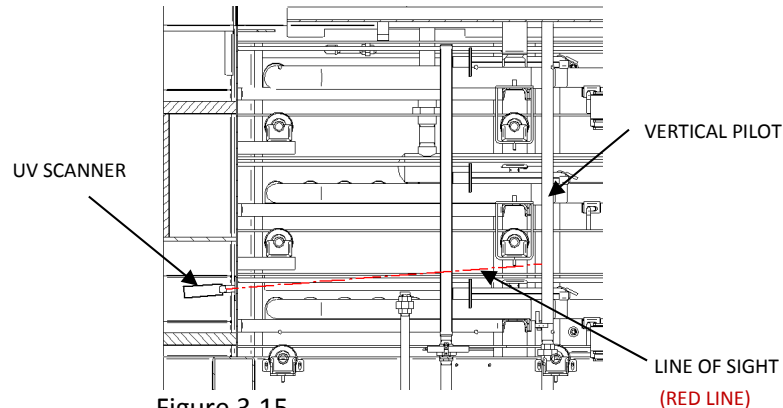


Figure 3.15

5. If there is a solid flame on the pilot, and the main burners are still not lighting, please check the following:

5A. That there is a signal at the flame safety controller that signals a 'Flame' present. Hold a lighter in front of the U.V. scanner (Figure 3.16). If the red light next to 'Flame' is not registering, check that there is 120 volts AC at the U.V. scanner. If there is, then you must replace the U.V. scanner (Figure 3.14 & 3.15).



Figure 3.16

5B. If upon replacement of the U.V. scanner, the flame safety controller still does not see the pilot tree flame nor the lighter flame held in front and main gas valves do not open, then replace the 'Amplifier' in the Flame Safety Controller (Figure 3.17).



Figure 3.17

Tension on the Slat belts

1. To maintain proper tension on the Slat belts of the oven, you will need a measuring tape and a large adjustable wrench.
2. The tension is controlled by springs. These springs are connected by some all-thread bolts that are the same length. Each deck will have a left and a right side spring for tensioning. Each deck has an all-thread bolt that connects itself to the take-up bearing.
3. Using the measuring tape, measure the distance from the outside end or edge of the spring to the end of the all-thread bolt (Figure 3.18 & 3.19). These measurements must be exactly the same on the left side versus the right side springs.



Figure 3.18



Figure 3.19

4. When the oven is hot and the belts are turning, measure and adjust the tension. The proper tension will reduce or nearly eliminate any loud noises or bouncing of the slat belt as it changes direction.
5. If there is too much tension, the slat belt will run stiffly and bounce hard. This will cause uneven wear in the support rail, prematurely stretch the slat belts' chains, wear out the bearings sooner than normal and create loud operating noise.

6. If there is not enough tension from the springs, the chains of the slat belts will not disengage from the gears' teeth. The slat belts will chatter and bounce due to the extra sag in the belts. Every once in a while, you will get a violent release of the chain from the gears and also may hang onto the teeth too long. This will cause severe damage if the chain does not release from the gear teeth. A properly tensioned slat belt that is lubricated daily should not run loudly during production.

Lubrication of the slat belt chains

1. The slat belt chains need to be lubricated every shift for quiet, smooth and long lasting operation. The recommended oil for lubricating the chains has to be 'Food Grade' and be able to handle temperatures of 400 – 700 degrees Fahrenheit.
2. Casa Herrera recommends TL-650 chain oil. It is a food grade, high temperature lube that will not smoke nor carbonize with use.
3. This chain oil is applied with a drip bottle or brush when the oven belts are cold enough to touch. This is done for safety of the operator/maintenance staff during the application of the lubricant.

Oven bearing lubrication

1. The oven bearings must be lubricated every 4 – 7 days depending on how many shifts the oven runs without stopping. High temperature grease is recommended. The oven bearings are critical for long lasting belt life, shaft life and slat belt life. The types of grease that can be used are listed in the manual under 'Lubricants'.

CHINS 106

Drive Roller chains...

THE GREASE

Use a high quality "Cling Grease". This grease has the ability to cling to the steel sprockets and chains even at high room temperatures and relatively high running speeds. Your local lubrication supplier will recommend the appropriate grease part number for this application. It is available in food grade as well as non-food grade.

APPLICATION

Brush onto entire chain on initial application. Run for several minutes and wipe off any grease which has been thrown from the drive.

MAINTENANCE

Once every week

Wipe excess grease from the chain and sprockets, check sprocket and chain condition and tensioning, and re-brush with cling grease as noted above

Once every 180 days

Remove the chain from the drive and clean in a suitable and compatible solvent. Dry the chain, reinstall and re-lubricate as noted above.

Notation 1

If a sprocket is badly worn, replace both the worn sprocket and the chain at the same time. Running a worn sprocket will destroy the chain and visa versa.

Notation 2

Do not wash the chain, bearings shafts, or sprockets with caustic or acid solutions under any circumstance. Use a proper solvent.

OVEN LUBRICATION

WARNING!

TURN OFF MACHINERY BEFORE LUBRICATING EQUIPMENT. LUBRICATING MOVING MACHINERY IS VERY DANGEROUS. USE SAFETY LOCKOUTS TO PREVENT OPERATION. USE CAUTION AT ALL TIMES.

CAUTION!

USE ONLY THE LUBRICANTS ASSIGNED BY THE MAINTENANCE SUPERVISOR, AND ONLY IN THE PRESCRIBED WAY. DUE TO DIFFERENCES IN CHEMISTRY, SOME OILS AND GREASES WILL DESTROY EACH OTHER WHEN THEY ARE MIXED TOGETHER. IF A CHANGE IS NEEDED, IT IS NECESSARY TO COMPLETELY FLUSH OUT THE OLD LUBRICANT BEFORE THE NEW IS PUT IN ITS PLACE.

1. Lubricate all pillow block and flange bearings, using grease approved by maintenance supervision for that purpose. Lubricate sparingly. Do not rupture the seals!
2. Lube all bearings once a week through fittings provided. Do not force excess lube through seals.
3. Use a high-quality grease. Consult the chart in this section for recommended lubricants.
4. Clean the excess grease thoroughly.
5. Be sure not to miss any bearings in the drive train underneath the sheeter head, or at the discharge end of the oven.
6. Lubricate roller-chains and sprockets with approved chain oil. NOTE: This instruction does NOT apply to drive chains that are a built-in-part of slat baking belts.



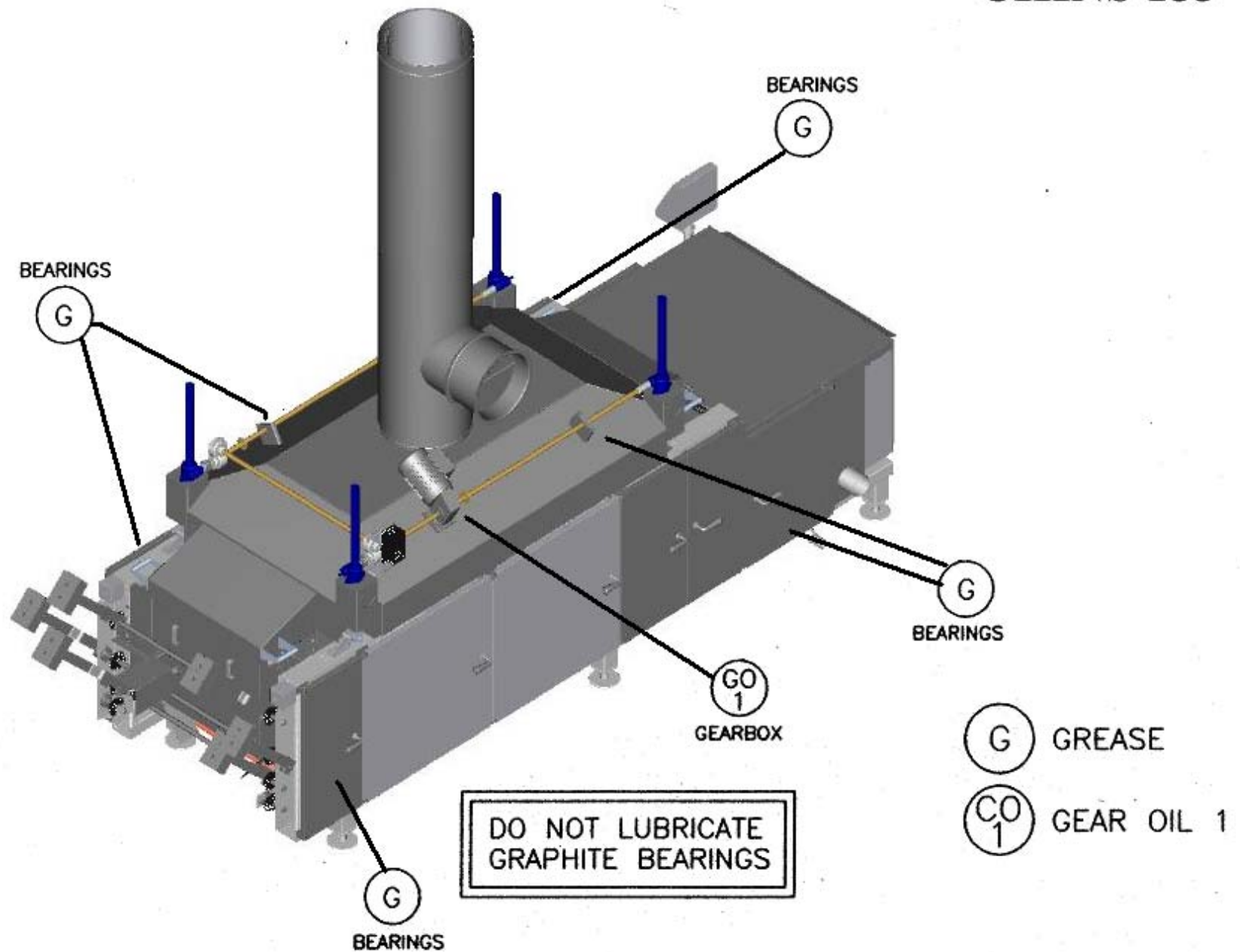
7. SLAT BELTS WITH ROLLER-CHAIN DRIVES: apply Graphite Lube DAILY after production. Prepare lubricant as follows:
 - a. Add three parts Kerosene to one part Graphite.
 - b. Mix Thoroughly
 - c. Using a 3" brush, spread this solution generously on the BELT ROLLER CHAIN ONLY, NOT ON SLATS!
8. Check speed reducer gear box each week for leaks.
Flush and replace oil yearly: See lube chart and MINS sheets for reducer data.
9. For location of Major Lubrication Points see diagram in this section.

AND WHILE YOU'RE AT IT.....

Check the condition of sprockets and chains for signs of abnormal wear.

Check alignment of sprockets.

Check idlers to see whether the chain is too slack, loose, or tight.



MAJOR LUBRICATION POINTS, OVEN

CHINS 105

LUBRICATION CHART

	 Grease	 Grease	 GEAR OIL 1			 GEAR OIL 2		 CHAIN OIL	 HYDRAULIC OIL
			16 - 50°F	51 - 110°F	111 - 165°F	5 - 77°F	32 - 104°F		
American Oil						Spartan EP-150	Spartan EP-220		
Atlantic Refining									
Cities Service			CITGO Cylinder Oil 460-5	CITGO Cylinder Oil 680-7	CITGO Cylinder Oil 680-7				
Chevron						Non-Leaded Gear Compound 150	Non-Leaded Gear Compound 220		
Dixon	Bak-Pruf EG #2 HP Water White	HY Heat GP Graphite Grease							
Gulf Oil			Transgear EP-460	Transgear EP-680	Transgear EP-800	EP Lube S-60	EP Lube S-100		
Keystone			WG-A	K-600	WG-Special				
Mobil Oil			Mobil 600W	Mobil 600W Super	Mobil Extra HECLA	Mobilgear 629	Mobilgear 630		DTE 25 Food Grade
Shell Oil			Omala 460	Omala 680	Omala 800	Moata Oil 100	Omata Oil 220		
Standard Oil									
Sun Oil			Sunep 1090	Sunep 1150	Sunoco Gear Oil .8 AC				
Texaco			Merota 220	Vanguard 620	Vanguard 620	Merota 150	Merota 220	Waylube 68	Randa HD46 Food Grade
Valvoline	Valpex EP2		Light Cylinder Oil	Light Cylinder Oil	Light Cylinder Oil	HP Gear Oil 80W-85-90	HP Gear Oil 80W-85-90	Waylube W-100	